

Basic Principles of Medicine II

Neuroscience Practice Questions

Lecture:

- **Neurotransmitter Pathways and Roles**

Questions Only

NB 09 – Neurotransmitter Pathways and Roles

1. A 33-year-old man is brought to the emergency department because of a 12-hour history of muscle spasms. He has a 48-hour history of a deep penetrating wound on the bottom of his left foot. Laboratory studies show tetanus toxaemia. Release of which of the following transmitters is most directly altered by the implicated toxin?

A. Glycine
B. Norepinephrine
C. Acetylcholine
D. Glutamate
2. A 63-year-old man comes to the physician because of a 5-day history of anxiousness. His anxiety is such that he is unable to perform normal household tasks. A short course of drug therapy is prescribed. Which of the following receptor complexes is most likely targeted by the drug?

A. Glutamate
B. GABA
C. Glycine
D. Dopamine
E. Norepinephrine
F. Acetylcholine
3. A 78-year-old man comes to the physician because of a 3-week history of uncontrollable shaking of his hands, slowing of movements like walking, and muscle stiffness. A diagnosis of Parkinson's disease is made

following a physical examination. Which of the following neurotransmitters is most likely affected in this patient?

- A. Acetylcholine
- B. Dopamine
- C. GABA
- D. Glutamate
- E. Norepinephrine

4. A 40-year-old woman is brought to the physician because she sees things no one else sees and hears voices no one else hears. She is also paranoid and says she is being tracked. Examination findings and laboratory studies are consistent with a psychotic episode. She is subsequently given antipsychotic medications. Which of the following is the most likely feature of the administered medication?

- A. Inhibits glycine
- B. Stimulates glycine
- C. Inhibits dopamine
- D. Stimulates dopamine
- E. Inhibits glutamate
- F. Stimulates glutamate

5. A 45-year-old woman with severe anxiety is being treated with a benzodiazepine due to its role in augmenting GABAergic transmission. Which of the following enzymes is most likely responsible for the conversion of glutamate to GABA?

- A. GABA transaminase
- B. Glutaminase
- C. Glutamic acid decarboxylase
- D. Glutamine synthase

6. A 70-year-old man is brought to the physician by his wife because of a 1-year history of memory problems. His wife reports that her husband's memory has been declining steadily and she no longer feels confident leaving him alone in the house. She says he regularly forgets where he is and what he is supposed to be doing. More recently, he failed to

recognize his daughter. The patient is not oriented to place or time. Mental status examination shows deficits in short- and long-term memory. He does not show any insight into his memory issues. Pathological changes to which of the following neurotransmitter systems most likely underlie the patient's deficits?

- A. Acetylcholine
- B. Dopamine
- C. GABA
- D. Glutamate
- E. Serotonin

7. In an experimental setting investigating arousal and attention, neurons in the locus ceruleus are examined. Which neurotransmitter is most likely associated with these neurons?

- A. Acetylcholine
- B. Dopamine
- C. Serotonin
- D. Norepinephrine

Questions with Answers and Explanations

NB 09 – Neurotransmitter Pathways and Roles

1. A 33-year-old man is brought to the emergency department because of a 12-hour history of muscle spasms. He has a 48-hour history of a deep penetrating wound on the bottom of his left foot. Laboratory studies show tetanus toxaemia. Release of which of the following transmitters is most directly altered by the implicated toxin?

A. Glycine
B. Norepinephrine
C. Acetylcholine
D. Glutamate

Answer Key: A

Acetylcholine release onto the neuromuscular junction would be elevated in this patient. However, this increase would be secondary to the disinhibition of lower motor neurons. Primarily, tetanus toxin reduces the spinal and bulbar release of glycine, thereby leading to disinhibition of the lower motor neurons and the onset of rigor.

2. A 63-year-old man comes to the physician because of a 5-day history of anxiousness. His anxiety is such that he is unable to perform normal household tasks. A short course of drug therapy is prescribed. Which of the following receptor complexes is most likely targeted by the drug?

A. Glutamate
B. GABA
C. Glycine
D. Dopamine
E. Norepinephrine
F. Acetylcholine

Answer Key: B

Benzodiazepines often exert anxiolytic effects. The relevant mechanism of action is the allosteric binding of the drug to its recognition site on the GABA-A receptor complex with resultant enhancement of chloride influx.

3. A 78-year-old man comes to the physician because of a 3-week history of uncontrollable shaking of his hands, slowing of movements like walking, and muscle stiffness. A diagnosis of Parkinson's disease is made following a physical examination. Which of the following neurotransmitters is most likely affected in this patient?

- A. Acetylcholine
- B. Dopamine
- C. GABA
- D. Glutamate
- E. Norepinephrine

Answer Key: B

Parkinson's disease is a neurodegenerative disease characterized by diminished dopamine levels due to the loss of dopaminergic neurons in the substantia nigra.

4. A 40-year-old woman is brought to the physician because she sees things no one else sees and hears voices no one else hears. She is also paranoid and says she is being tracked. Examination findings and laboratory studies are consistent with a psychotic episode. She is subsequently given antipsychotic medications. Which of the following is the most likely feature of the administered medication?

- A. Inhibits glycine
- B. Stimulates glycine
- C. Inhibits dopamine
- D. Stimulates dopamine
- E. Inhibits glutamate
- F. Stimulates glutamate

Answer Key: C

Due to increased dopamine activity present in patients with psychosis, the mechanism of action of antipsychotics involves blocking dopamine.

5. A 45-year-old woman with severe anxiety is being treated with a benzodiazepine due to its role in augmenting GABAergic transmission.

Which of the following enzymes is most likely responsible for the conversion of glutamate to GABA?

- A. GABA transaminase
- B. Glutaminase
- C. Glutamic acid decarboxylase
- D. Glutamine synthase

Answer Key: C

Glutamic acid decarboxylase converts glutamate to GABA. GABA transaminase is involved in the conversion of GABA to glutamate.

Glutaminase – glutamine to glutamate. Glutamine synthase – glutamate to glutamine

6. A 70-year-old man is brought to the physician by his wife because of a 1-year history of memory problems. His wife reports that her husband's memory has been declining steadily, and she no longer feels confident leaving him alone in the house. She says he regularly forgets where he is and what he is supposed to be doing. More recently, he failed to recognize his daughter. The patient is not oriented to place or time. Mental status examination shows deficits in short- and long-term memory. He does not show any insight into his memory issues. Pathological changes to which of the following neurotransmitter systems most likely underlie the patient's deficits?

- A. Acetylcholine
- B. Dopamine
- C. GABA
- D. Glutamate
- E. Serotonin

Answer Key: A

This elderly patient shows significant, chronic, worsening memory issues consistent with abnormalities in basal forebrain cholinergic neurotransmission as seen in Alzheimer's disease.

7. In an experimental setting investigating arousal and attention, neurons in the locus coeruleus are examined. Which neurotransmitter is most likely associated with these neurons?

- A. Acetylcholine
- B. Dopamine
- C. Serotonin
- D. Norepinephrine

Answer Key: D

The locus coeruleus, located in the pons, contains an aggregation of noradrenergic neurons. They send projections to the thalamus, hypothalamus, limbic structures, and cerebral cortex

- A. There are two groups of cholinergic neurons in the CNS: the basal forebrain group, which includes the basal nucleus of Meynert, and the dorsolateral pontine tegmentum group.
- B. Dopaminergic neurons are mainly found in the substantia nigra, ventral tegmental area, and the hypothalamic arcuate nucleus.
- C. Serotonin-containing neurons are situated in the raphe nuclei throughout the brainstem.